

PHARMA LMS

Admin Portal — 360° User Stories

Complete Feature Specifications, Event Workflows & Acceptance Criteria
Enterprise Grade · Sun Pharma Standard · 21 CFR Part 11 · EU GMP Annex 11 · ICH Q10

Attribute	Detail
Total Modules	14 functional modules across the full Admin Portal
Total User Stories	92 user stories covering every admin capability
Compliance Standards	21 CFR Part 11 · EU GMP Annex 11 · ALCOA+ · GxP · ICH Q10 · GAMP 5
Target Organisation	Enterprise pharmaceutical (Sun Pharma scale — 40,000+ employees)
Portal Stack	Flutter/Dart frontend · Serverpod backend · PostgreSQL · Redis · Kafka · S3
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1 — User & Identity Management

Covers: creating/deactivating user accounts, assigning roles and departments, SSO/LDAP federation, multi-factor authentication policy, password policy enforcement, and privileged access reviews.

Regulatory basis: 21 CFR 11.10(g) — authority checks; 21 CFR 11.300 — code-based identification.

US-ADM-USR-001

As a **System Administrator**, I want to create a new user account with role, department, and site assignment, so that new employees can access the correct portal and courses immediately upon hire

Acceptance Criteria

Required fields: first name, last name, employee ID, email, department, site, role (EMPLOYEE/TRAINER/ADMIN/AUDITOR)

Employee ID is validated against HR system sync table (or manually entered)

Default password generated and emailed; must be changed on first login

If email domain matches SSO config, SSO flag is auto-set

Audit event USER_CREATED logged with my admin ID, timestamp, IP

User appears in Employee Portal within 5 seconds of creation

Event Workflow

- 1 Admin fills creation form — validates employee ID uniqueness
- 2 System checks HR sync table for matching record
- 3 Password hash generated (PBKDF2-SHA512); welcome email queued to Kafka topic user.created
- 4 INSERT INTO users, user_roles, user_department_assignments (single transaction)
- 5 INSERT INTO audit_trail (action:USER_CREATED, entity:user, new_value:JSONB snapshot)

6 Kafka consumer sends welcome email with temporary credentials

US-ADM-USR-002

As a **System Administrator**, I want to bulk-import users from a CSV or via HR system integration (SAP/Workday), so that onboarding an entire batch of new hires or a site acquisition does not require manual entry

Acceptance Criteria

CSV template: employee_id, first_name, last_name, email, department, site, role, manager_id

HR sync pulls from HR system API on schedule (daily) or on-demand trigger

Conflict resolution rules: if employee_id exists --> UPDATE, else INSERT

Import preview shows exact changes before commit (new: N, updated: U, unchanged: --)

Errors (invalid dept codes, duplicate emails) shown row-by-row; partial import allowed

Full import log downloadable as CSV; retained 7 years for audit

Event Workflow

1 Admin triggers import (CSV upload or HR API pull)

2 System validates each row against schema + referential integrity

3 Validation report generated (N valid / E errors / U updates)

4 Admin reviews report and confirms or rejects

5 On confirm: batch INSERT/UPDATE inside transaction; Kafka events per new user

6 Audit trail: BULK_USER_IMPORT event with row counts and file hash

US-ADM-USR-003

As a System Administrator , I want to assign and modify user roles using a granular permission model, so that the principle of least privilege is enforced and every portal section is gated correctly
Acceptance Criteria
Roles: EMPLOYEE, TRAINER, MANAGER, QA_REVIEWER, AUDITOR, CONTENT_ADMIN, SUPER_ADMIN
Roles can be stacked (e.g. TRAINER + QA_REVIEWER)
Role change takes effect immediately on next API call (JWT re-issued on next login)
Previous role is retained in role_history table (append-only)
SUPER_ADMIN role requires dual approval (two-person rule) before assignment
All role changes audit-logged with old_value, new_value, changed_by
Event Workflow
1 Admin selects user and chooses "Modify Roles"
2 System shows current roles and permission matrix preview
3 If SUPER_ADMIN being assigned: workflow triggers second-approver email
4 Second approver logs in, reviews, approves or rejects
5 On approval: UPDATE user_roles; INSERT role_history; INSERT audit_trail
6 JWT cache invalidated for that user (Redis DEL user:{id}:session)
US-ADM-USR-004
As a System Administrator , I want to deactivate or terminate a user account with full data preservation, so that departing employees can no longer access the system while all training records are retained indefinitely for regulatory inspection
Acceptance Criteria
Deactivation is a soft delete: active=false; NO data deleted

All active sessions revoked immediately (Redis flush for user)
All pending enrollments flagged SUSPENDED (not deleted)
Certificates remain valid and searchable by auditors
Reason for deactivation required (resignation, transfer, termination)
Account cannot be fully deleted if user has any audit_trail records
Reactivation is possible (restores enrollments if within 90 days)
Event Workflow
1 Admin opens user → clicks "Deactivate"
2 System shows impact: N active sessions, M pending enrollments
3 Admin enters deactivation reason (GxP reason field)
4 Transaction: UPDATE users SET active=false; DELETE user_sessions; UPDATE enrollment_records SET status=SUSPENDED
5 Kafka: user deactivated topic notifies downstream systems (HR, badge system)
6 INSERT audit_trail {USER_DEACTIVATED, old_value:{active:true}, new_value:{active:false,reason}}
US-ADM-USR-005
As a System Administrator , I want to configure SSO / LDAP integration so employees log in with corporate credentials, so that password management is centralised in the corporate IdP (Okta / Azure AD / ADFS) and users do not maintain separate LMS passwords
Acceptance Criteria
SSO configuration: SAML 2.0 or OIDC provider URL, entity ID, certificate
LDAP: server URL, bind DN, base DN, attribute mapping (email → email, department → dept)
Test connection button verifies connectivity before saving

Fallback: if SSO unavailable, local password allowed for ADMIN and AUDITOR roles only
JIT provisioning: new SSO user auto-creates LMS account with EMPLOYEE role on first login
Group-to-role mapping: AD group → LMS role (configurable)
Event Workflow
1 Admin navigates to System → SSO Configuration
2 Enters IdP metadata URL or uploads XML
3 Maps SAML attributes to LMS user fields
4 Clicks "Test SSO" → system performs SP-initiated auth flow in sandbox
5 On success: UPDATE system_config SET sso_enabled=true, sso_provider_config=JSONB
6 INSERT audit_trail (SSO_CONFIG_UPDATED, changed_by, timestamp)
US-ADM-USR-006
As a System Administrator , I want to enforce password and MFA policies system-wide, so that authentication meets 21 CFR 11.300 requirements for code-based electronic signatures
Acceptance Criteria
Password policy fields: min length, complexity, expiry days, reuse prevention (last N)
MFA policy: enforce for roles ADMIN/AUDITOR/OA_REVIEWER (configurable)
MFA methods: TOTP (Google Authenticator), email OTP, hardware token (FIDO2)
Session timeout: idle timeout (minutes) and absolute timeout (hours)
Failed login lockout threshold and lockout duration configurable
Policy changes take effect at next login; grace period configurable
Event Workflow

- 1 Admin sets policy in System → Security Policy
- 2 Validates: min_length ≥ 8, expiry ≤ 365, lockout ≥ 3 attempts
- 3 UPDATE security_policy table (single row, versioned)
- 4 INSERT audit_trail (SECURITY_POLICY_UPDATED, old_value, new_value)
- 5 Background job: flag all accounts whose passwords now violate new policy → mark password_expired=true
- 6 Users forced to change password on next login

US-ADM-USR-007

As a **System Administrator**, I want to manage organisational hierarchy: departments, sites, and reporting lines, so that training assignments, compliance reports, and escalation rules can be scoped correctly to the organisational structure

Acceptance Criteria

- Organisation tree: Company → Site → Division → Department → Team
- Departments have: code, name, head (FK→users), compliance_category, gmp_area flag
- Sites have: name, country, regulatory_region (US, EU, APAC)
- Department head receives escalation notifications for overdue staff
- Moving a department changes all unresolved enrollment scoping in real time

Event Workflow

- 1 Admin opens Organisation Management
- 2 Creates or edits department node in tree
- 3 UPDATE departments: cascade update to users.department_id if dept renamed
- 4 Kafka: org structure changed triggers re-evaluation of batch membership rules

5 INSERT audit_trail (ORG_STRUCTURE_UPDATED)

US-ADM-USR-008

As a **System Administrator**, I want to **conduct a quarterly privileged access review and recertify or revoke elevated roles**, so that we maintain continuous compliance with principle of least privilege and can produce access review evidence for FDA/EMA inspections

Acceptance Criteria

System generates access review report: all users with ADMIN, AUDITOR, QA_REVIEWER roles

Report shows: user, role, date granted, last login, granting admin

Admin must approve (recertify) or revoke each record within review window

Unreviewed records auto-revoke at window close (configurable)

Review result stored in access_reviews table with digital signature

Report is exportable as signed PDF for regulatory submission

Event Workflow

1 Scheduled job triggers ACCESS_REVIEW_INITIATED on first day of review quarter

2 Admin receives review task; opens Access Review screen

3 For each user: Admin clicks Recertify or Revoke

4 Revoke triggers deactivation workflow (see US-ADM-USR-004)

5 Recertify: INSERT access_reviews (user_id, role, decision:APPROVED, reviewed_by, signed_at)

6 On window close: all pending = auto-revoked; INSERT audit_trail (ACCESS_REVIEW_COMPLETED)

2 — Course & Content Management

Covers: creating courses, uploading SCORM/xAPI/PDF content, versioning, approval workflows, content catalogues, course retirement, and linking courses to SOPs and regulatory standards.

Regulatory basis: GMP Annex 11 §10 — software validation; GAMP 5 category classification.

US-ADM-CRS-001

As a Content Administrator, I want to create a new course with full metadata and regulatory classification, so that every course is traceable to a regulatory requirement and can be included in inspection-ready training records

Acceptance Criteria

Course fields: title, code, version, description, learning objectives, department(s), duration_minutes

Regulatory fields: gmp_category, regulation_reference (e.g. 21 CFR 211.68), gamp5_class

Settings: pass_threshold (%), max_attempts, cert_validity_days, requires_e_signature_on_completion

Auto-generated course code: COURSE-{DEPT_CODE}-{SEQ}

Course starts in DRAFT status; must go through approval workflow before ACTIVE

Course code + version together must be unique

Event Workflow

- 1 Admin fills course creation form
- 2 System validates: unique code+version, valid dept codes, pass_threshold 1–100
- 3 INSERT INTO courses (status=DRAFT)
- 4 INSERT audit_trail {COURSE_CREATED, entity:course, new_value:JSONB}
- 5 Admin assigned as course owner; approval request auto-sent to designated reviewer

US-ADM-CRS-002

As a Content Administrator, I want to upload SCORM 1.2 / SCORM 2004 / xAPI (Tin Can) packages and validate them before publishing, so that learners receive a consistent, validated learning experience and we have SCORM API compatibility evidence on file

Acceptance Criteria

Upload accepted: .zip SCORM packages up to 500 MB

System auto-detects SCORM version from imsmanifest.xml

Validation checks: imsmanifest.xml present, all referenced files exist, no broken internal links

Validation report shows pass/fail per check with fix guidance

SCORM package stored in S3/MinIO; SHA-256 hash recorded for integrity

Preview mode available before publishing

Event Workflow

- 1 Admin uploads SCORM zip to course edit screen
- 2 S3Service.uploadScorm() stores zip; extracts manifest to parse metadata
- 3 ScormValidationService runs 12-point checklist; returns validation_result JSONB
- 4 INSERT scorm_packages {course_id, s3_key, file_hash, scorm_version, validation_result}
- 5 UPDATE courses SET scorm_package_id=? if validation passed
- 6 INSERT audit_trail {SCORM_UPLOADED, file_hash, validation_result}

US-ADM-CRS-003

As a Content Administrator, I want to manage course versions with an immutable change history, so that learners who completed version 1.x have a permanent record showing which version they completed, and new learners always get the current approved version

Acceptance Criteria

New version = new courses row with incremented version_number; previous row status → SUPERSEDED

In-progress enrollments on old version: admin chooses (complete on old version OR migrate to new)

New version must go through full approval workflow before replacing current

Version diff view shows what changed between versions

Change reason required for every new version (GxP)

Superseded versions remain searchable and reportable

Event Workflow

- 1 Admin opens course → "Create New Version"
- 2 System clones course record with version+1, status=DRAFT
- 3 Admin makes changes, uploads new SCORM if applicable
- 4 On publish approval: UPDATE old course SET status=SUPERSEDED, superseded_at=NOW()
- 5 UPDATE courses SET status=ACTIVE for new version
- 6 Kafka: course.version.superseded → triggers notification to open-enrollment learners

US-ADM-CRS-004

As a Content Administrator, I want to configure a multi-step course approval workflow before any course goes live, so that every course published to employees has been reviewed by a Subject Matter Expert and a QA reviewer, meeting GAMP 5 validation documentation requirements

Acceptance Criteria

Workflow stages: Content Review (SME) → QA Review → Final Approval (Head of Training)

Each stage has an assigned reviewer role; can be a specific user or any user with that role

Reviewer can: Approve, Reject (with mandatory comments), or Return for Revision

Stage timeout: reviewer has N days before escalation email

Full workflow history stored in course_approval_workflow table

Approved course auto-transitions to ACTIVE; rejected returns to DRAFT

Event Workflow

- 1 Admin submits course for review → creates workflow_instance record**
- 2 Kafka: course.review.requested → email to Stage 1 reviewer**
- 3 Reviewer opens course, reviews content, submits decision with comments**
- 4 On each stage approval: UPDATE workflow_instance SET stage=next; notify next reviewer**
- 5 On final approval: UPDATE courses SET status=ACTIVE, approved_by, approved_at**
- 6 INSERT audit_trail {COURSE_APPROVED, workflow_instance_id, all_stage_decisions JSONB}**

US-ADM-CRS-005

As a Content Administrator, I want to manage a searchable course catalogue with tagging and prerequisite chains, so that admins can build structured learning paths and employees can discover relevant content

Acceptance Criteria

Catalogue supports: categories, tags, regulatory references, prerequisite courses

Prerequisite enforcement: learner cannot start Course B until Course A is COMPLETED

Admin can set: recommended sequence, estimated total path duration

Learning paths: ordered sequence of courses with a single enrollment action

Search: full-text search on title, description, tags, regulation_reference

Catalogue visible to employees as read-only; assignment triggers enrollment

Event Workflow

- 1 Admin creates catalogue entry or tags existing course
- 2 For prerequisites: INSERT course_prerequisites {course_id, prerequisite_course_id, is_mandatory}
- 3 For learning paths: INSERT learning_paths; INSERT learning_path_courses (ordered)
- 4 Employee portal reads catalogue via GET /api/v1/catalogue with full-text search
- 5 Prerequisite check runs on every enrollment attempt (server-side)

US-ADM-CRS-006

As a Content Administrator, I want to retire or archive a course that is no longer required, so that outdated content does not appear in active catalogues and no new enrollments are created, while all historical records are preserved

Acceptance Criteria

Retirement requires: retirement reason, effective date, replacement course (optional)

Cannot retire a course with active enrollments unless migrated or manually closed

Retired course: status=RETIRED; removed from catalogue; no new enrollments possible

Historical completions, certificates, and audit events remain fully queryable

Retirement must be approved by Head of Training (workflow)

Event Workflow

- 1 Admin initiates retirement → system checks active enrollment count
- 2 If active enrollments > 0: admin must choose migrate/close action per enrollment
- 3 After resolution: UPDATE courses SET status=RETIRED, retired_at, retirement_reason

4 Kafka: course.retired → remove from catalogue index; notify enrolled learners

5 INSERT audit_trail {COURSE_RETIRED}

US-ADM-CRS-007

As a Content Administrator, I want to link courses to specific SOPs, regulations, and competency frameworks, so that our training records demonstrate that each course satisfies a specific regulatory or quality requirement — essential for FDA 483 responses

Acceptance Criteria

Each course can link to: sop_versions (many-to-many), regulation references (text), competency_types

Link shows: SOP number, SOP version, effective date

If an SOP is superseded, linked courses are flagged for review

Competency mapping: completing course grants/contributes to a competency_type level

Traceability matrix: SOP → Training requirement → Course → Completion records

Event Workflow

1 Admin opens course → "Regulatory Links" tab

2 Selects SOP versions and regulation references from controlled picklists

3 INSERT course_sop_links {course_id, sop_version_id}

4 INSERT course_regulation_links {course_id, regulation_ref}

5 Traceability report queries these join tables + enrollment_records

3 — Enrollment Management

Covers: individual enrollment, bulk CSV enrollment, rule-based auto-enrollment, due date management, enrollment modification, cancellation, re-enrollment after failure, and compliance tracking.

Regulatory basis: 21 CFR 11.10(e) — audit trails for training; ALCOA+ attribution.

US-ADM-ENR-001

As a LMS Administrator, I want to enroll a single employee in one or more courses with custom due dates, so that targeted training can be assigned within seconds for new hires, role changes, or remedial action

Acceptance Criteria

- Employee search: by name, employee ID, email, department
- Course selection: from catalogue; multiple courses allowed in one action
- Due date: custom (overrides course default) or course default
- Reason field required (GxP audit trail)
- System prevents duplicate active enrollment (same user+course+version)
- Employee notified by email and in-app notification on enrollment
- Audit event ENROLLMENT_CREATED logged with admin ID

Event Workflow

- Admin searches and selects employee + courses
- System runs duplicate check: SELECT FROM enrollment_records WHERE user_id=? AND course_id=? AND status NOT IN (CANCELLED,SUPERSEDED)
- On confirm: INSERT enrollment_records (one row per course)
- INSERT audit_trail {ENROLLMENT_CREATED} per enrollment

5 Kafka: training.enrollment.created → employee notification consumer

US-ADM-ENR-002

As a LMS Administrator, I want to view and filter all enrollments with live status across the entire organisation, so that I have a single source of truth for training compliance status that I can present during an FDA inspection or internal audit

Acceptance Criteria

Table columns: employee, course, department, status, progress %, due date, enrolled date, assigned by

OVERDUE status shown in red (#DC2626) at all times — non-negotiable

Filters: status, department, site, course, due date range, assigned by, compliance_required

Real-time counts: total, overdue, in-progress, completed, enrolled

Export filtered view to CSV (export event audit-logged)

Clicking a row opens enrollment detail with full session history and audit timeline

Event Workflow

1 Admin opens Enrollments → applies filters

2 Backend: SELECT with dynamic WHERE clauses; pagination (LIMIT/OFFSET)

3 Overdue logic: WHERE due_date < NOW() AND status NOT IN (COMPLETED, CANCELLED)

4 Export: server generates CSV stream; INSERT audit_trail {ENROLLMENT_LIST_EXPORTED}

US-ADM-ENR-003

As a LMS Administrator, I want to bulk-enroll employees using a CSV upload with row-level validation, so that an entire cohort (hundreds of employees) can be enrolled in minutes for a regulatory training cycle

Acceptance Criteria

CSV columns: employee_id, course_code, course_version, due_date, reason, compliance_required, priority

Max rows per upload: 5,000

Validation: employee exists, course active, due date valid, no duplicate enrollment

Preview report before commit: N valid, E errors, W warnings (duplicates)

Partial import: skip error rows, enroll valid rows

Import result log downloadable; retained in system as import_job record

All created enrollments trigger standard notifications

Event Workflow

- 1 Admin uploads CSV → POST /api/v1/enrollments/bulk-validate
- 2 Validation service parses CSV, checks each row against DB constraints
- 3 Returns validation_report JSONB: rows array with status per row
- 4 Admin confirms → POST /api/v1/enrollments/bulk-commit with job_id
- 5 Background job: INSERT enrollment_records in batches of 100
- 6 INSERT import_jobs {job_id, admin_id, file_hash, rows_valid, rows_error, rows_skipped}
- 7 INSERT audit_trail {BULK_ENROLLMENT_IMPORTED, job_id, counts}

US-ADM-ENR-004

As a LMS Administrator, I want to configure rule-based auto-enrollment triggers so courses are assigned automatically based on HR events, so that new hires get their mandatory training immediately on Day 1 without manual admin intervention, supporting our zero-gap onboarding SLA

Acceptance Criteria

Rules: ON_HIRE, ON_TRANSFER (department change), ON_PROMOTION, ANNUAL_RENEWAL
Each rule: trigger_event, target_group (dept/role/job_spec), course_list, due_offset_days
Rules can have conditions: e.g. "only if site = India" or "only if grade ≥ L3"
Test mode: simulate rule against current user population to preview impact
Conflicting rules: admin resolves via priority ordering
Event Workflow
1 Admin creates auto-enrollment rule in Rules Engine
2 INSERT enrollment_rules {trigger, conditions JSONB, courses JSONB, due_offset}
3 HR integration (or manual trigger) fires Kafka event: hr.employee.onboarded
4 Rules engine consumer evaluates all rules against the event payload
5 Matching rules: INSERT enrollment_records with source=AUTO_RULE, rule_id
6 INSERT audit_trail {AUTO_ENROLLMENT_TRIGGERED, rule_id, user_id, courses}
US-ADM-ENR-005
As a LMS Administrator, I want to modify an enrollment: extend due date, upgrade course version, or change priority, so that I can respond to legitimate training plan changes (course updates, project delays) with a full paper trail
Acceptance Criteria
Due date extension: requires reason; logged as ENROLLMENT_DUE_DATE_CHANGED
Version upgrade: migrates learner from superseded to new version; session progress may carry over
Priority change: affects reminder scheduling
Each modification: old_value and new_value both stored in audit trail

Completed enrollments: cannot modify due date retroactively (immutable)
Event Workflow
1 Admin opens enrollment detail → clicks Edit
2 Selects field to change; enters new value and mandatory reason
3 UPDATE enrollment_records SET {field}=? WHERE id=? AND status != COMPLETED
4 INSERT audit_trail {ENROLLMENT_MODIFIED, old_value, new_value, reason, changed_by}
5 If version upgrade: UPDATE enrollment_records SET course_id=new_version_id
US-ADM-ENR-006
As a LMS Administrator, I want to cancel an enrollment and document the reason, so that enrollments that are no longer valid (employee left, course restructured) are cleanly closed without data loss
Acceptance Criteria
Cancellation requires mandatory reason (GxP)
Completed enrollments cannot be cancelled (only archived)
In-progress sessions are force-closed with status CANCELLED
Certificates (if any) linked to the cancelled enrollment remain valid
Cancelled enrollment retained in DB (soft delete only)
Event Workflow
1 Admin opens enrollment → Manage → Cancel
2 System warns: "employee has N active sessions"
3 Admin enters cancellation reason
4 UPDATE enrollment_records SET status=CANCELLED, cancelled_at, cancelled_by, cancellation_reason

5 UPDATE training_sessions SET status=ABANDONED for any IN_PROGRESS sessions

6 INSERT audit_trail {ENROLLMENT_CANCELLED, reason}

7 Kafka: training.enrollment.cancelled → employee notification

US-ADM-ENR-007

As a LMS Administrator, I want to grant a re-enrollment after a failed assessment attempt, resetting the attempt counter, so that an employee who legitimately failed (e.g. technical issues, extenuating circumstances) can re-attempt without losing their training record

Acceptance Criteria

Re-enrollment creates a NEW enrollment_records row (preserves failed one)

Previous attempts remain in assessment_attempts (immutable)

Re-enrollment requires: reason, approver (manager or QA)

Attempt counter reset on new enrollment row

Original failure record linked to new enrollment via re_enrollment_of FK

Event Workflow

1 Admin opens failed enrollment → "Grant Re-enrollment"

2 Enters reason; selects approver if required

3 INSERT enrollment_records {same user+course, status=ENROLLED, re_enrollment_of=original_id}

4 INSERT audit_trail {RE_ENROLLMENT_GRANTED, original_enrollment_id, reason, granted_by}

5 Employee notified of re-enrollment opportunity

US-ADM-ENR-008

As a LMS Administrator, I want to view a per-employee training transcript showing all historical enrollments and completions, so that I can produce a complete training history for any employee in seconds during an audit or when responding to a regulatory inquiry

Acceptance Criteria

Transcript: all enrollments ever, with status, dates, scores, certificates, e-signature records

Chronological view with filters by year, course category, status

Shows SUPERSEDED course completions with version clearly labelled

Export as PDF (formatted, signed with timestamp) or CSV

PDF export event is audit-logged (TRANSCRIPT_EXPORTED)

Event Workflow

- 1 Admin opens user → Training Transcript tab
- 2 Backend: SELECT all enrollment_records for user_id including CANCELLED and SUPERSEDED
- 3 JOIN training_sessions, certificates, e_signatures
- 4 PDF export: server-side PDF generation with digital watermark and admin signature
- 5 INSERT audit_trail {TRANSCRIPT_EXPORTED, exported_by, export_format}

4 — Batch & Cohort Management

Covers: creating named training batches (cohort + courses + timeline), activating batches, monitoring batch progress, adding/removing members, cloning batches for recurring cycles, and batch-level compliance reporting.

US-ADM-BAT-001

As a LMS Administrator, I want to create a named training batch with departments, courses, and a shared timeline, so that a full training cycle (e.g. annual GMP refresher) can be managed as a single unit rather than hundreds of individual enrollments

Acceptance Criteria

Batch fields: name, description, start_date, end_date, regulatory_category, departments, courses, priority

Per-course due offset (days from batch start date) configurable

Members: resolved by department selection, job spec, or individual search

Batch saved as DRAFT before activation; no enrollments created until activation

Batch code auto-generated: BATCH-{YEAR}-{SEQ}

Status lifecycle: DRAFT → ACTIVE → COMPLETING → COMPLETED → ARCHIVED

Event Workflow

- 1 Admin fills batch creation form
- 2 INSERT training_batches {status=DRAFT, metadata JSONB}
- 3 INSERT batch_courses per course
- 4 INSERT batch_member_rules (department or job_spec based)
- 5 INSERT audit_trail {BATCH_CREATED}

US-ADM-BAT-002

As a LMS Administrator, I want to activate a batch, which atomically creates all enrollment records for all members and courses, so that hundreds of enrollments are created in a single auditable transaction with no risk of partial state

Acceptance Criteria

Activation preview: shows exact member count, course count, total enrollment rows to be created

Duplicate check: existing active enrollments for same user+course are skipped (logged)

All enrollments created in a single DB transaction

Employee notifications sent via Kafka after transaction commits

Batch status transitions DRAFT → ACTIVE

Activation locked for 24h after creation (prevents accidental immediate activation)

Event Workflow

- 1 Admin clicks "Activate Batch" → preview modal shows impact
- 2 On confirm: BEGIN TRANSACTION
- 3 Resolve members: `SELECT users WHERE department IN batch.departments AND active=true`
- 4 For each member × course: `INSERT enrollment_records` (skip if duplicate)
- 5 `UPDATE training_batches SET status=ACTIVE, activated_at, activated_by`
- 6 COMMIT TRANSACTION
- 7 `INSERT audit_trail {BATCH_ACTIVATED, member_count, enrollment_count, skipped_count}`
- 8 Kafka: `training.batch.activated` → fan-out notification job

US-ADM-BAT-003

As a LMS Administrator, I want to monitor batch progress in real time with a roster view and per-course completion gauges, so that I can identify at-risk members early and intervene before the batch deadline

Acceptance Criteria

Dashboard: overall completion %, per-course completion %, overdue member count

Roster table: each member with per-course status (Enrolled / In Progress / Completed / Overdue / Failed)

Sort by: overall completion %, overdue count, last activity date

Filter: overdue-only, completed-only, specific course incomplete

One-click "Send reminder to all overdue members in this batch"

Batch completion date estimated based on current pace

Event Workflow

- 1 Admin opens batch detail
- 2 Backend: complex query — GROUP BY user_id, course_id over enrollment_records + training_sessions
- 3 Overdue detection: WHERE due_date < NOW() AND status NOT IN (COMPLETED)
- 4 Reminder action: Kafka: batch.reminder.requested → notification job for all overdue in batch
- 5 INSERT audit_trail {BATCH_REMINDER_SENT, batch_id, member_count}

US-ADM-BAT-004

As a LMS Administrator, I want to add or remove members from an active batch with an audit trail, so that staff changes (new hires joining mid-batch, transfers out) are reflected in the batch without disrupting other members

Acceptance Criteria

Add member: creates enrollment_records for all batch courses (same logic as activation)

Remove member: cancels their enrollments in this batch (not other enrollments)
Reason required for both add and remove
Members added mid-batch: due dates calculated from add_date + course offset (not batch start)
Event Workflow
1 Admin opens batch roster → Add/Remove
2 On add: INSERT enrollment_records for new_member × batch_courses
3 INSERT batch_members {batch_id, user_id, added_at, added_by, add_reason}
4 On remove: UPDATE enrollment_records SET status=CANCELLED for batch scope
5 INSERT audit_trail {BATCH_MEMBER_ADDED / BATCH_MEMBER_REMOVED, reason}
US-ADM-BAT-005
As a LMS Administrator, I want to clone an existing batch to create a new training cycle with refreshed dates and re-resolved membership, so that annual or quarterly recurring training cycles can be set up in under 5 minutes without rebuilding from scratch
Acceptance Criteria
Clone copies: name (appended suffix e.g. "— Q2 2026"), courses, department filters, due offsets, notification settings
Clone does NOT copy: members (re-resolved at activation), completion records, previous batch data
Cloned batch starts in DRAFT with new start/end date fields blank (must be set)
Membership is re-resolved at activation to pick up new hires since last cycle
Event Workflow
1 Admin opens batch → Clone
2 System creates new training_batches row with cloned_from_batch_id, status=DRAFT

3 Clones batch_courses and batch_member_rules

4 INSERT audit_trail {BATCH_CLONED, source_batch_id, new_batch_id}

US-ADM-BAT-006

As a LMS Administrator, I want to generate a batch completion report for QA submission or regulatory filing, so that I can produce a compliance artefact showing every member's training completion within a batch, suitable for inclusion in a Periodic Quality Review or audit response

Acceptance Criteria

Report: batch metadata, member list, per-member per-course status, scores, completion dates, certificate IDs

Summary statistics: overall %, per-course %, pass rate

PDF generated with digital watermark (batch ID, generated by, timestamp)

CSV variant for data analysis

Report generation is audit-logged

Event Workflow

1 Admin clicks "Generate Report" on batch detail

2 Backend assembles report dataset from enrollment_records, training_sessions, certificates

3 Server-side PDF generated (headless rendering)

4 Stored in S3 with report_key; URL returned to admin

5 INSERT batch_reports {batch_id, report_s3_key, generated_by, generated_at}

6 INSERT audit_trail {BATCH_REPORT_GENERATED}

5 — Job Specifications & Training Matrix

Covers: creating job specifications (role training profiles), training matrix management, assigning courses to employees by job role, compliance gap analysis per role, and job spec versioning for GMP change control.

US-ADM-JOB-001

As a LMS Administrator, I want to create and manage job specifications that define required and optional training for each job role, so that every employee hired or transferred can be enrolled in all role-appropriate training with one action, driven by a single maintained source of truth

Acceptance Criteria

Fields: title, code, department, grade_range, regulatory_category, description, effective_date

HR system integration: link spec to one or more HR job codes for auto-assignment

Course entries: course, requirement_type (REQUIRED/RECOMMENDED/N_A), due_offset_days, renewal_months

Versioned: every edit creates new version; old version immutable (GMP change control)

Approval workflow before publishing (SME + Head of Training sign-off)

Event Workflow

1 Admin fills job spec form with courses and requirement levels

2 INSERT job_specifications {status=DRAFT}

3 INSERT job_spec_courses per course entry

4 On submit for approval: workflow triggers approver email

5 On approval: UPDATE job_specifications SET status=PUBLISHED

6 INSERT audit_trail {JOB_SPEC_PUBLISHED, spec_id, version, approved_by}

US-ADM-JOB-002

As a LMS Administrator, I want to assign a job specification to an employee and auto-create all required enrollments, so that new hires and transfers get their full training plan created instantly with correct due dates calculated from their start date

Acceptance Criteria

- System resolves required courses from job spec
- Optional courses shown as pre-selected checkboxes (admin can deselect)
- Due dates: employee start_date + spec.due_offset_days per course
- Duplicate enrollments skipped automatically
- Preview: "X courses will be enrolled; Y already enrolled"

Event Workflow

- 1 Admin selects employee + job spec
- 2 Backend resolves courses from job_spec_courses WHERE requirement_type = REQUIRED
- 3 For each course: INSERT enrollment_records {source=JOB_SPEC, job_spec_id}
- 4 INSERT audit_trail {JOB_SPEC_ASSIGNED_TO_USER, spec_id, user_id, enrollments_created}
- 5 Kafka: training.bulk.assigned → notification fan-out

US-ADM-JOB-003

As a LMS Administrator, I want to view and edit the training matrix — a role × course grid showing required/optional/N-A status, so that I can audit our entire training coverage in one view and identify gaps before a regulatory inspection

Acceptance Criteria

Matrix rows: all active job specs; columns: all active courses
Cell values: R (Required), O (Optional), — (Not Applicable)
Inline cell editing: click cell to toggle R/O/—
Filter by department, regulatory category
Export: Excel (for regulatory submission) or PDF
Change to any cell creates a new job_spec version (GMP change control)
Event Workflow
1 Admin opens Training Matrix
2 Backend: SELECT job_spec_courses JOIN job_specifications JOIN courses
3 Pivot in application layer to produce role × course grid
4 Cell edit: UPDATE job_spec_courses SET requirement_type=? → triggers new spec version
5 INSERT audit_trail {TRAINING_MATRIX_UPDATED, cell: {spec_id, course_id, old_val, new_val}}
US-ADM-JOB-004
As a LMS Administrator, I want to run a compliance gap analysis against a job specification to identify employees missing required training, so that I can close training gaps proactively before they become findings in an FDA inspection
Acceptance Criteria
Gap: employee assigned to job spec but missing required course completion or certificate
Report shows: employee, spec, missing course, gap type (never enrolled / enrolled but overdue / cert expired)
One-click enroll all gapped employees in missing courses
Report filterable by department, site, urgency

Event Workflow
1 Admin selects job spec → Run Gap Analysis
2 Query: users with job_spec_id JOIN required courses LEFT JOIN completions WHERE completed IS NULL OR cert expired
3 Gap records returned; admin reviews
4 Bulk enroll action: INSERT enrollment_records for all gap records
5 INSERT audit_trail {GAP_ANALYSIS_RUN, spec_id, gaps_found, enrollments_created}
US-ADM-JOB-005
As a LMS Administrator, I want to manage a job specification renewal schedule to ensure certifications are refreshed on time, so that employees never inadvertently fall out of compliance due to certificate expiry because the system auto-enrolls them before their certificate expires
Acceptance Criteria
renewal_months on each spec course drives auto-re-enrollment
System calculates renewal_due_date = certificate.expires_at - lead_time_days (configurable)
Auto-enrollment rule: when renewal_due_date approaches, enroll employee in refresher course
Admin can view: all upcoming renewals in next 90 days
Bulk enroll: enroll all employees with renewals due in date range
Event Workflow
1 Scheduled job runs daily: SELECT certificates where expires_at BETWEEN NOW() AND NOW()+lead_time
2 For each expiring cert: check if employee already has active enrollment in refresher
3 If not: INSERT enrollment_records {source=RENEWAL_AUTO, expiring_cert_id}
4 Kafka: certificate.renewal.enrollment.created

5 INSERT audit_trail {RENEWAL_ENROLLMENT_CREATED, cert_id, user_id}

US-ADM-JOB-006

As a LMS Administrator, I want to produce a per-role compliance report showing overall training compliance % for each job specification, so that department heads and QA managers can see at a glance which job roles are fully trained and which are at risk, supporting the Periodic Quality Review process

Acceptance Criteria

Report: job spec × compliance metrics (% fully compliant, % partial, % non-compliant)

Drill down: per-employee compliance against the spec

Time-series: compliance trend over last 12 months

Export for QA submission

Event Workflow

1 Admin generates report from Job Spec detail or Reports section

2 Query aggregates enrollment_records, certificates per job_spec_id

3 Compliance logic: all REQUIRED courses COMPLETED + valid cert (or N/A)

4 Results rendered in dashboard and as downloadable PDF/CSV

5 INSERT audit_trail {COMPLIANCE_REPORT_GENERATED, spec_id}

US-ADM-JOB-007

As a LMS Administrator, I want to version-control job specifications with a documented change history and change reason, so that every change to the training requirements for a role is traceable — who changed what, when, and why — meeting GMP change control documentation requirements

Acceptance Criteria

Every edit: admin must enter change_reason before saving
New version created (old version status → SUPERSEDED)
Version diff view: side-by-side comparison of course list changes
Superseded version remains fully searchable
A user can have been assessed against a specific spec version (linked via enrollment)
Event Workflow
1 Admin edits published spec → enters change_reason
2 System clones spec row with version+1, status=DRAFT
3 Approval workflow runs
4 On approval: UPDATE old spec SET status=SUPERSEDED; UPDATE new spec SET status=PUBLISHED
5 INSERT job_spec_versions {spec_id, version, changed_by, change_reason, diff_json}
6 INSERT audit_trail {JOB_SPEC_NEW_VERSION, old_version, new_version, reason}

6 — Assessment Management

Covers: question bank management, assessment creation and configuration, scoring rules, attempt policies, randomisation, proctoring settings, result review, and re-attempt policies.

Regulatory basis: 21 CFR 11.10(e) — immutable assessment records; ALCOA+ accuracy.

US-ADM-ASS-001

As a Content Administrator, I want to build and manage a question bank with categorised, versioned questions, so that assessments can pull from a controlled pool of validated questions without duplication, and individual question changes are tracked

Acceptance Criteria

Question types: single choice, multiple choice, true/false, short text, scenario-based

Each question: text, type, category, difficulty, explanation, regulation_reference

Questions are versioned; editing creates new version (old version immutable if used in published assessment)

Question status: DRAFT → REVIEWED → APPROVED → ACTIVE

Bulk import: CSV or XML (QTI format)

Event Workflow

1 Admin creates/edits question → INSERT assessment_questions {status=DRAFT}

2 Submit for review → same approval workflow as courses

3 On approval: UPDATE assessment_questions SET status=ACTIVE

4 INSERT audit_trail {QUESTION_APPROVED, question_id, version}

US-ADM-ASS-002

As a Content Administrator, I want to create an assessment linked to a course with configurable scoring and randomisation, so that every course completion can require demonstrated knowledge, not just content consumption

Acceptance Criteria

Assessment fields: title, course_id, passing_score (%), max_attempts, time_limit_minutes, randomise_questions (bool), randomise_options (bool)

Question selection: pick specific questions OR define pool rules (e.g. 10 random from category X)

Must pass assessment: if true, enrollment cannot complete until assessment passed

Re-attempt cooldown: hours/days between attempts (configurable)

Event Workflow

- 1 Admin creates assessment and links questions
- 2 INSERT assessments {course_id, config JSONB}
- 3 INSERT assessment_question_pool entries
- 4 INSERT audit_trail {ASSESSMENT_CREATED}

US-ADM-ASS-003

As a LMS Administrator, I want to review individual assessment attempt records including every submitted answer, so that in case of dispute or audit, I can show exactly what answers were submitted, when, from which IP, and the score calculation

Acceptance Criteria

View: attempt metadata (start, submit, duration, IP, device)

View: each question → selected answer → correct answer → points awarded

Cannot modify attempt records (immutable per 21 CFR 11)

Can annotate attempt with admin note (appended, never edits the original)

Event Workflow
1 Admin opens assessment attempt from enrollment detail
2 SELECT assessment_attempts JOIN assessment_responses JOIN assessment_questions
3 All data read-only; annotation: INSERT attempt_annotations {attempt_id, note, annotated_by, annotated_at}
4 INSERT audit_trail {ASSESSMENT_ATTEMPT_VIEWED, attempt_id, viewed_by}
US-ADM-ASS-004
As a LMS Administrator, I want to override an assessment result in exceptional circumstances (technical failure, question error), so that a legitimate learner is not penalised by a system error, while the override is fully documented and auditable
Acceptance Criteria
Override requires: reason, authorisation_level (SUPER_ADMIN only)
Two-person rule: override requested by one admin, approved by another
Original attempt record is never modified; override creates separate record
Override record: original_attempt_id, override_score, override_reason, requested_by, approved_by, signed_at
Event Workflow
1 Admin requests override → INSERT assessment_overrides {status=PENDING}
2 Second approver reviews and approves
3 On approval: system evaluates if override score meets passing_threshold
4 If yes: INSERT certificates (if not already issued)
5 INSERT audit_trail {ASSESSMENT_OVERRIDE_APPLIED, original_attempt_id, override_id, both_signatories}

US-ADM-ASS-005

As a LMS Administrator, I want to configure and enforce a minimum time requirement per assessment, so that employees cannot complete assessments instantaneously (gaming), maintaining the integrity of the competency verification process

Acceptance Criteria

min_time_seconds configurable per assessment

Enforced server-side: submission rejected if (submitted_at - started_at) < min_time

Client-side timer shows remaining minimum time before submission enabled

Attempt_at_minimum_time_violation event logged if bypassed via API

Event Workflow

- 1 Server receives submit request: checks (NOW() - attempt.started_at) vs min_time
- 2 If violation: UPDATE assessment_attempts SET status=INVALID, violation_type=MIN_TIME
- 3 INSERT audit_trail {ASSESSMENT_MIN_TIME_VIOLATION, attempt_id}

US-ADM-ASS-006

As a Content Administrator, I want to configure assessment proctoring settings including camera, screen recording, and browser lockdown, so that regulated assessments that qualify employees for controlled operations maintain the integrity required for FDA-inspectable competency verification

Acceptance Criteria

Proctoring levels: NONE, HONOUR (declaration only), MONITORED (webcam), STRICT (screen + webcam)

Browser lockdown: prevent copy-paste, tab switching, external links

For MONITORED/STRICT: webcam capture stored in S3 for N days (configurable)

Admin can review proctoring footage for flagged attempts

Event Workflow

1 Admin configures proctoring_level on assessment
2 On learner start: client initialises proctoring mode based on level
3 STRICT: browser API locks down tab; webcam stream uploaded to S3 as chunked video
4 Violation events (tab switch, camera blocked) logged to proctoring_events table
US-ADM-ASS-007
As a LMS Administrator, I want to generate an assessment analytics report showing score distributions, question difficulty, and pass rates by department, so that I can identify poorly performing questions, departments with low pass rates, and systemic training gaps requiring content revision
Acceptance Criteria
Metrics: pass rate (overall and by dept), average score, score distribution histogram, per-question difficulty index
Question difficulty: % of attempts that answered correctly
Trend: pass rate over time (6-month trend)
Filter by course, department, date range
Event Workflow
1 Complex aggregation query over assessment_attempts, assessment_responses
2 GROUP BY assessment_id, question_id, department
3 Difficulty index: SUM(is_correct) / COUNT(*) per question
4 Results rendered in analytics dashboard + exportable

7 — Certificate Management

Covers: certificate template management, auto-issuance on completion, manual issuance, expiry and renewal notifications, certificate revocation, verification, and bulk certificate reporting.

Regulatory basis: 21 CFR 11.50 — e-signature on certificates; 21 CFR 11.10(e).

US-ADM-CRT-001

As a LMS Administrator, I want to manage certificate templates with organisation branding, digital signature blocks, and custom fields, so that every certificate issued to employees is professionally formatted and contains all required regulatory information

Acceptance Criteria

Template fields: title, header logo, footer text, signature_blocks (up to 3), custom_fields (course, date, validity)

Signature blocks: title, name, date (auto-populated on issuance)

Templates versioned; changing template does not retroactively change issued certificates

Preview mode: generates sample PDF

Event Workflow

- 1 Admin creates/edits template in Certificate Template editor
- 2 INSERT certificate_templates {template_data JSONB, version}
- 3 INSERT audit_trail {CERTIFICATE_TEMPLATE_UPDATED}

US-ADM-CRT-002

As a LMS Administrator, I want to configure auto-issuance rules: which courses generate certificates and under what conditions, so that certificate issuance is systematic and consistent, not dependent on manual admin action

Acceptance Criteria

Conditions: assessment passed (if applicable) AND SCORM completed AND e-signature (if required)

Certificate linked to specific course version (employee always knows which version they completed)

Validity period: configured per course (in months); NULL = no expiry

Auto-generate PDF and store in S3 on issuance

Event Workflow

- 1 Completion event triggers: `TrainingService.checkCertificateConditions()`
- 2 If all conditions met: `CertificateService.generate()`
- 3 INSERT certificates {cert_hash=SHA256(user_id+course_id+session_id+issued_at)}
- 4 PDF generated server-side; stored in S3 cert_key
- 5 INSERT audit_trail {CERTIFICATE_ISSUED, cert_id, conditions_met JSONB}

US-ADM-CRT-003

As a LMS Administrator, I want to manually issue a certificate to an employee (e.g. for legacy training or equivalent experience recognition), so that employees with prior training that predates the LMS or from external programmes can have their competency recognised in the system

Acceptance Criteria

Requires: course, employee, completion_date, justification, supporting document upload

Must be approved by QA Reviewer (workflow)

Manual certificate flagged with issuance_type=MANUAL for easy identification in audits

Event Workflow

- 1 Admin fills manual certificate form + uploads evidence
- 2 INSERT certificates {issuance_type=MANUAL, status=PENDING_APPROVAL}

3 Approval workflow triggers QA reviewer email

4 On approval: UPDATE certificates SET status=ACTIVE; INSERT audit_trail {CERTIFICATE_MANUALLY_ISSUED}

US-ADM-CRT-004

As a LMS Administrator, I want to revoke a certificate with a documented reason and maintain the revocation record, so that certificates issued in error or where the employee has demonstrated non-compliance can be invalidated, with full traceability for inspectors

Acceptance Criteria

Revocation requires: reason, approving admin (two-person rule)

Revoked certificate status = REVOKED; remains in DB

Employee notified of revocation

Certificate PDF marked VOID (watermark overlay)

Any active enrollments for renewal of this cert are also cancelled

Event Workflow

1 Admin initiates revocation → INSERT certificate_revocation_requests

2 Second approver approves

3 UPDATE certificates SET status=REVOKED, revoked_at, revoked_by, revocation_reason

4 PDF watermarking: S3 PUT with VOID overlay

5 Kafka: certificate.revoked → employee notification

6 INSERT audit_trail {CERTIFICATE_REVOKED, both_signatories, reason}

US-ADM-CRT-005

As a LMS Administrator, I want to monitor certificate expiry across the organisation and trigger renewal enrollments, so that no employee falls out of compliance due to an expired certificate without being enrolled in a refresher course

Acceptance Criteria

- Dashboard: certificates expiring in 0–30, 31–60, 61–90 days
- Sorted by: criticality (regulatory-required certs first)
- Bulk action: enroll all employees with expiring certs in the designated refresher course
- Auto-renewal rules (see Job Spec module) can handle this automatically

Event Workflow

- 1 Admin opens Certificate Expiry dashboard
- 2 `SELECT certificates WHERE expires_at < NOW()+90 AND status=ACTIVE`
- 3 Bulk enroll action: `INSERT enrollment_records` for selected employees
- 4 `INSERT audit_trail {RENEWAL_ENROLLMENT_BULK_CREATED, count}`

US-ADM-CRT-006

As a LMS Administrator, I want to generate and export a bulk certificate register for regulatory submission, so that I can provide auditors with a complete, timestamped list of all active, expired, and revoked certificates in any date range

Acceptance Criteria

- Register: employee, course, cert number, issued_at, expires_at, status, cert_hash
- Filter by: status, department, site, course, date range
- Exportable as signed PDF (timestamped, admin signature) or CSV

Event Workflow

- 1 Admin applies filters → Generate Register

2 SELECT certificates JOIN users JOIN courses with filter predicates

3 PDF rendered with digital signature block

4 INSERT audit_trail {CERTIFICATE_REGISTER_EXPORTED, filter_params, row_count}

8 — Document & SOP Management

Covers: controlled document upload and versioning, SOP assignment to departments, acknowledgement workflows, e-signature collection, supersession management, and read receipts.

Regulatory basis: 21 CFR 11.10(b) — accurate copies; GMP Annex 11 §14 — document control.

US-ADM-DOC-001

As a Document Control Administrator, I want to upload a new controlled document (SOP, policy, work instruction) with full metadata, **so that** every document in the LMS has a clear owner, a regulatory classification, and an effective date — providing an inspection-ready document management trail

Acceptance Criteria

Fields: title, document_number, version, type (SOP/WI/Policy/Form), department(s), effective_date, review_date, owner, regulatory_reference

PDF uploaded to S3/MinIO; SHA-256 file hash recorded for integrity

Document goes through approval workflow before becoming ACTIVE

Supersedes field: link to previous version if applicable

Event Workflow

- 1 Admin uploads PDF + fills metadata form
- 2 S3Service.uploadDocument() → S3 PUT; returns s3_key
- 3 INSERT sop_versions {metadata, s3_key, file_hash, status=DRAFT}
- 4 Approval workflow triggered
- 5 INSERT audit_trail {DOCUMENT_UPLOADED, sop_version_id, file_hash}

US-ADM-DOC-002

As a Document Control Administrator, I want to manage the document approval workflow with configurable review stages, so that no document reaches employees without going through the required QA/regulatory review process

Acceptance Criteria

Workflow stages: Author Review → Technical Review → QA Approval → Effective

Each stage: assigned reviewer role, deadline, approve/reject/return actions

Full workflow history stored with comments

On final approval: document becomes ACTIVE; employees can view

Event Workflow

- 1 Author submits for review → INSERT document_workflow_instances
- 2 Each stage approval: UPDATE workflow_instance SET stage; notify next reviewer
- 3 On final approval: UPDATE sop_versions SET status=ACTIVE, approved_by, approved_at
- 4 INSERT audit_trail {DOCUMENT_APPROVED, all_stage_decisions JSONB}

US-ADM-DOC-003

As a Document Control Administrator, I want to assign documents to departments and define which employees must acknowledge them, so that only relevant employees see each document, and required acknowledgements are tracked for compliance

Acceptance Criteria

Assignment: document → department(s) or specific job specs

Acknowledgement required: YES/NO per assignment

Due date for acknowledgement configurable

When a new document is assigned, employees notified immediately

Event Workflow
1 Admin sets department assignments
2 INSERT department_sop_assignments {sop_version_id, department_id, requires_ack, ack_due_date}
3 Kafka: document.assigned → notification to all employees in assigned departments
4 INSERT audit_trail {DOCUMENT_ASSIGNED, departments, requires_ack}
US-ADM-DOC-004
As a Document Control Administrator, I want to track acknowledgement completion across all assigned employees and chase non-completers, so that I can demonstrate 100% document awareness compliance across every department before an inspection
Acceptance Criteria
Dashboard: document → acknowledgement % per department
OVERDUE acknowledgements shown in red (#DC2626)
Bulk reminder: send reminder to all employees who have not acknowledged by due date
Acknowledgement records: employee, timestamp, IP, e-signature reference
Event Workflow
1 SELECT sop_acknowledgements LEFT JOIN department employees to find missing
2 Overdue: WHERE ack_due_date < NOW() AND acknowledged_at IS NULL
3 Reminder Kafka event: document.acknowledgement.reminder
4 INSERT audit_trail {DOCUMENT_REMINDER_SENT, document_id, recipient_count}
US-ADM-DOC-005

As a Document Control Administrator, I want to supersede an old document version with a new one and force re-acknowledgement from all assigned employees, so that when an SOP is updated, all employees are immediately aware of the new version and their old acknowledgement is no longer valid

Acceptance Criteria

Supersession: new version becomes ACTIVE; old version status → SUPERSEDED

Existing acknowledgements of old version are NOT transferred to new version

Employees with old version acknowledgement automatically receive new version for re-acknowledgement

Admin configures grace period for re-acknowledgement (days)

Event Workflow

- 1 Admin publishes new version → links supersedes_version_id**
- 2 UPDATE old sop_version SET status=SUPERSEDED, superseded_at**
- 3 UPDATE new sop_version SET status=ACTIVE**
- 4 Background job: identify all employees with old version assignment → create new acknowledgement_required records**
- 5 Kafka: document.superseded → notifications**
- 6 INSERT audit_trail {DOCUMENT_SUPERSEDED, old_version_id, new_version_id}**

US-ADM-DOC-006

As a Document Control Administrator, I want to generate a document read and acknowledgement report for regulatory submission, so that I can produce evidence that every employee in a GMP area has read and understood the latest version of every required SOP — a core FDA inspection requirement

Acceptance Criteria

Report: per-document listing of all assigned employees with acknowledgement status and date

Summary: total assigned, acknowledged, pending, overdue

Filter by department, document, date range
Export as signed PDF with digital watermark
Event Workflow
1 Complex query: department_sop_assignments JOIN users LEFT JOIN sop_acknowledgements
2 PDF generated server-side with digital signature
3 INSERT audit_trail {DOCUMENT_ACKNOWLEDGEMENT_REPORT_EXPORTED}
US-ADM-DOC-007
As a Document Control Administrator, I want to link training courses to specific SOPs so training gaps are automatically identified when an SOP is updated, so that when QA updates SOP-QA-002, the system can immediately identify all enrolled employees who may need refresher training on the updated procedures
Acceptance Criteria
SOP-to-course link: sop_version → course(s)
When SOP is superseded: linked courses are flagged for review
Admin can trigger auto-enrollment in linked refresher course for all employees in the SOP department
Event Workflow
1 Admin sets course_sop_links from document detail
2 On SOP supersession: background job queries course_sop_links
3 Flags linked courses with needs_review=true
4 Admin receives notification: "SOP-QA-002 was updated — review linked courses"
US-ADM-DOC-008

As a Document Control Administrator, I want to configure document retention policies and archive documents past their retention period, so that we comply with FDA data retention requirements (minimum 3 years for training records, longer for product-specific SOPs) while managing storage costs

Acceptance Criteria

Retention policy: by document_type, duration_years (3, 5, 7, 10, permanent)

Documents past retention: auto-archived (read-only, not deleted) after admin confirmation

Archive confirmation requires e-signature

Archived documents still searchable by auditors

Event Workflow

- 1 Scheduled job: SELECT documents WHERE effective_date < NOW()-retention_years*365
- 2 INSERT archival_queue for each eligible document
- 3 Admin reviews queue → confirms archival with e-signature
- 4 UPDATE sop_versions SET status=ARCHIVED, archived_at
- 5 INSERT audit_trail {DOCUMENT_ARCHIVED, retention_policy_applied, e_signature_id}

9 — Compliance & Gap Reporting

Covers: compliance dashboards, real-time gap detection, department-level reports,

regulatory readiness reports, trend analysis, and scheduled report delivery.

Regulatory basis: 21 CFR 11.10(e) — audit trails available: GMP Annex 11 §9 — reporting.

US-ADM-RPT-001

As a **Senior Compliance Manager**, I want to view a real-time compliance dashboard showing training status across the entire organisation, so that I always have an up-to-date view of our compliance posture and can answer an auditor question in under 60 seconds

Acceptance Criteria

KPIs: overall compliance %, overdue count, overdue % by department, certificates expiring in 30 days

Status breakdown: enrolled, in-progress, completed, overdue, failed

Site comparison: side-by-side compliance % per manufacturing site

Trend chart: weekly/monthly compliance trend over 12 months

Drill down: department → employee list

Event Workflow

- 1 Dashboard queries: multiple aggregation queries over enrollment records, certificates
- 2 Cached in Redis with 5-minute TTL for performance (10,000+ employees scale)
- 3 Cache invalidated on any enrollment status change (Kafka consumer)
- 4 INSERT audit_trail (COMPLIANCE_DASHBOARD_VIEWED)

US-ADM-RPT-002

As a **Department Head**, I want to generate a training compliance gap report for a specific department or site, so that department heads can act on gaps and I can produce documentary evidence for QA audits

Acceptance Criteria

Gap definition: employee has required course (per job spec or direct assignment) with status not COMPLETED

Gap types: NOT_ENROLLED, ENROLLED_OVERDUE, FAILED, CERT_EXPIRED

Report: employee * course * gap_type * days_since_due

One-click action: enroll all gapped employees in missing courses

Event Workflow

- 1 Query: users JOIN enrollment_rules/job_spec_courses to resolve all required courses
- 2 LEFT JOIN enrollment_records to identify missing or incomplete
- 3 Returns gap_records array
- 4 Export as CSV/PDF; INSERT audit_trail (GAP_REPORT_GENERATED)

US-ADM-RPT-003

As a **Department Head**, I want to generate an FDA/EMA inspection-ready training summary report, so that when a regulatory inspector asks for training records, I can produce a complete, formatted, digitally signed report in under 5 minutes

Acceptance Criteria

Report scope: all training completed by specified employees/departments in a date range

Includes: course name, version, completion date, score, certificate ID, e-signature ID

Digital signature: report signed with admin e-signature (21 CFR 11.50)

Report frozen: generated at a point in time; stored in S3 with hash for integrity

Event Workflow
1 Admin configures report parameters
2 Backend assembles dataset from enrollment_records, training_sessions, certificates, e_signatures
3 PDF generated with all required fields including cert hashes
4 Admin e-signature collected (re-authenticate required)
5 INSERT regulatory_reports (report_s3_key, file_hash, signed_by, signed_at, parameters)
6 INSERT audit_trail (REGULATORY_REPORT_GENERATED, e_signature_id)
US-ADM-RPT-004
As a System Administrator , I want to configure scheduled reports that are automatically generated and emailed to stakeholders , so that department heads, QA managers, and site directors receive regular training compliance updates without manual effort
Acceptance Criteria
Configurable: frequency (daily/weekly/monthly), recipient list, report type, filter scope
Recipients: specific users or users with a particular role
Report attached as PDF to email or available as dashboard link
Failure notification: if report generation fails, admin alerted
Event Workflow
1 Admin configures schedule → INSERT scheduled_reports (cron_expression, params, recipients)
2 Scheduled job fires: generates report, stores in S3
3 Kafka: report.generated → email consumer sends to recipients
4 INSERT audit_trail (SCHEDULED_REPORT_SENT, report_id, recipients)

US-ADM-RPT-005

As a **QA Manager**, I want to run a CAPA-linked compliance impact analysis when a corrective action requires training, so that when QA raises a CAPA that mandates retraining, I can immediately see which employees are affected and create targeted enrollments

Acceptance Criteria

- CAPA record: linked to specific courses or SOPs
- Impact analysis: employees in affected dept/role who need the linked training
- One-click: create targeted batch enrollment for CAPA-affected population
- CAPA progress tracked: % of affected employees trained

Event Workflow

- Admin creates CAPA record with course links
- System queries affected employees: dept/role filter
- Admin confirms → INSERT enrollment_records (source=CAPA, capa_id)
- CAPA compliance %: calculated from enrollment_records WHERE source=CAPA AND capa_id=?

US-ADM-RPT-006

As a **QA Manager**, I want to view and export a full regulatory traceability matrix linking SOPs to training courses to completion records, so that inspectors can follow the chain from a regulatory requirement → SOP → training course → employee completion in one report

Acceptance Criteria

- Matrix: Regulation → SOP version → Linked course → Employees who completed
- Filter by regulation type, department, date range

Export as Excel with navigation links between sections
Event Workflow
1 Complex JOIN: regulation_references → course_regulation_links → courses → enrollment_records
2 Multi-sheet Excel export generated server-side
3 INSERT audit_trail (TRACEABILITY_MATRIX_EXPORTED)
US-ADM-RPT-007
As a , I want to configure a compliance scoring methodology and view organisation-wide compliance health score, so that senior management and QA leadership have a single KPI for training compliance that rolls up across all departments and requirements
Acceptance Criteria
Score components: enrollment completion rate (40%), overdue resolution rate (30%), cert currency (20%), acknowledgement rate (10%)
Weighting configurable by admin
Score shown as 0–100; colour coded: >90 Green, 70-90 Amber, <70 Red
Trend: 12-month history of health score
Event Workflow
1 Scheduled job calculates score nightly
2 Stores in compliance_scores (score, component_scores JSONB, calculated_at)
3 Dashboard queries latest score + trend

10 — Audit Trail & CAPA Management

Covers: viewing and searching the immutable audit trail, HMAC chain integrity verification, raising and tracking CAPAs, deviation management, and regulatory inspection readiness.

Regulatory basis: 21 CFR 11.10(e) — computer-generated, computer-maintained records.

US-ADM-AUD-001

As a LMS Administrator / Auditor, I want to search and filter the immutable audit trail by user, event type, date range, and entity, so that I can answer any regulatory question about "who did what, when" within seconds during a live inspection

Acceptance Criteria

Search: user, event_type, entity_type, entity_id, date range, IP address

Results sorted: most recent first (default)

Each record shows: sequence_id, timestamp (UTC), user, action, entity, old_value, new_value, IP, HMAC hash

Records are read-only — no edit/delete buttons exist in UI

Export filtered results to CSV (export event itself is audit-logged)

Event Workflow

- 1 SELECT audit_trail with dynamic filter predicates
- 2 Pagination: LIMIT/OFFSET; max 1,000 rows per page
- 3 INSERT audit_trail {AUDIT_TRAIL_SEARCHED, filter_params, result_count, searched_by}

US-ADM-AUD-002

As a LMS Administrator, I want to run the HMAC chain integrity verification to confirm no audit trail records have been tampered with, so that I can demonstrate to an FDA inspector that our audit trail has not been altered since it was written

Acceptance Criteria

Verification algorithm: for each record in sequence order, recompute HMAC-SHA256(prev_hash + id + key_fields)

Compare computed hash to stored hmac_hash

Result: PASS (all records match) or FAIL (list of records with hash mismatch)

Verification run can be scheduled (monthly) or on-demand

Verification certificate: PDF with admin signature and timestamp

Event Workflow

- 1 Admin triggers verification → POST /api/v1/audit/verify-integrity
- 2 Server iterates audit_trail ORDER BY id; recomputes each hash
- 3 If mismatch found: INSERT integrity_violations {record_id, expected_hash, stored_hash}
- 4 Result stored in integrity_checks {run_by, run_at, records_checked, violations_found, status}
- 5 INSERT audit_trail {INTEGRITY_CHECK_RUN, result, violation_count}

US-ADM-AUD-003

As a LMS Administrator, I want to raise a CAPA (Corrective and Preventive Action) linked to a training failure or compliance gap, so that identified training deficiencies are formally tracked to resolution with ownership, timelines, and evidence — as required by ICH Q10 and GMP Annex 11

Acceptance Criteria

CAPA fields: title, description, root_cause, corrective_action, preventive_action, owner, due_date, linked_courses, linked_sops

CAPA status: OPEN → IN_PROGRESS → CLOSED_EFFECTIVE / CLOSED_INEFFECTIVE

CAPA closure requires evidence (document upload) and QA approval
Training actions within CAPA: create targeted enrollments (see Module 9)
Event Workflow
1 Admin raises CAPA → INSERT capa_records {status=OPEN}
2 Links to courses/SOPs → INSERT capa_links
3 CAPA owner notified
4 On closure: INSERT capa_closures {evidence_s3_key, closed_by, approved_by}
5 INSERT audit_trail {CAPA_RAISED / CAPA_CLOSED, capa_id}
US-ADM-AUD-004
As a LMS Administrator, I want to review all e-signature records and verify their integrity hashes, so that e-signatures on training completions, document acknowledgements, and assessments are cryptographically verified and tamper-evident
Acceptance Criteria
View: all e_signatures for a user, course, or date range
Each record: user, action, timestamp, integrity_hash
Integrity check: system recomputes HMAC-SHA256(signature_data+user_id+timestamp+secret)
Comparison: computed vs stored hash → VALID or TAMPERED flag
Event Workflow
1 SELECT e_signatures with filters
2 On-demand integrity check: ESignatureService.verify(signature_id)
3 INSERT audit_trail {ESIGNATURE_INTEGRITY_CHECKED, result}

US-ADM-AUD-005

As a LMS Administrator, I want to view a login and session audit report for any user, so that in response to a security incident or regulatory inquiry, I can reconstruct every login, session, and action taken by a specific user on a specific date

Acceptance Criteria

Report: all login events, session durations, IP addresses, devices, and actions taken per session

Shows: successful logins, failed attempts, account lockouts, MFA events

Chronological timeline view

Event Workflow

1 SELECT audit_trail WHERE user_id=? AND action_type IN (USER_LOGIN, USER_LOGOUT, ...)

2 JOIN user_sessions on session_id

3 INSERT audit_trail {USER_ACTIVITY_REPORT_VIEWED, target_user_id, viewed_by}

US-ADM-AUD-006

As a LMS Administrator, I want to prepare a complete regulatory inspection package on demand, so that when an FDA inspector arrives, I can produce a comprehensive, signed dossier of all training records, audit trails, and compliance evidence within 30 minutes

Acceptance Criteria

Inspection package: configurable scope (date range, departments, document types)

Contents: training summary, completion records, audit trail extract, certificate register, e-signature log, integrity check certificate

Package assembled as a ZIP archive with an index PDF

Admin e-signature required to finalise package (21 CFR 11.50)

Event Workflow

- 1 Admin configures package scope
- 2 System generates each component report in parallel
- 3 Assembles ZIP with index
- 4 Admin re-authenticates and applies e-signature
- 5 INSERT inspection_packages {scope_params, package_s3_key, signed_by, signed_at}
- 6 INSERT audit_trail {INSPECTION_PACKAGE_GENERATED, e_signature_id, package_id}

11 — Notifications & Communications

Covers: notification template management, automated reminder rules, escalation chains, broadcast announcements, and notification delivery tracking.

US-ADM-NOT-001

As a LMS Administrator, I want to configure notification templates for all system events with custom branding and content, so that all communications from the LMS reflect our organisation brand and contain the right information for the employee to take action

Acceptance Criteria

Template types: enrollment_created, enrollment_overdue, assessment_reminder, certificate_issued, sop_assigned

Template fields: subject, body (rich text with variables), priority

Variables: {{employee_name}}, {{course_name}}, {{due_date}}, {{portal_link}}

Multi-language templates supported

Event Workflow

- 1 Admin edits template → UPDATE notification_templates {body_html, subject, variables}
- 2 Test send: POST /api/v1/notifications/test with recipient=admin email
- 3 INSERT audit_trail {NOTIFICATION_TEMPLATE_UPDATED, template_type}

US-ADM-NOT-002

As a LMS Administrator, I want to configure automated reminder rules: when and how often employees are reminded about pending training, so that compliance rates improve through systematic nudging without requiring manual admin effort

Acceptance Criteria

Rules: trigger_type (days before due / days after due), frequency (once / daily / weekly), channel (email / in-app / both)

Escalation: if employee ignores after N reminders, notify manager

Quiet hours: configurable (do not send between 22:00–08:00 local time)

Reminder sent count tracked per enrollment

Event Workflow

- 1 Admin configures reminder_rules → INSERT reminder_rules {config JSONB}
- 2 Scheduled job evaluates rules against all active enrollments
- 3 For each matching enrollment: Kafka: notification.reminder.send
- 4 Consumer sends email/in-app; UPDATE enrollment_records SET reminder_count+1, last_reminder_at

US-ADM-NOT-003

As a LMS Administrator, I want to send a broadcast announcement to all employees or a specific department, so that I can communicate urgent compliance updates, system maintenance windows, or training deadlines to targeted groups

Acceptance Criteria

- Recipient scope:** All / Department / Site / Job Spec / Individual users
- Channel:** email, in-app notification, or both
- Scheduled send:** future date and time
- Read tracking:** which recipients have opened/dismissed the in-app notification

Event Workflow

- 1 Admin creates announcement → INSERT announcements {scope, content, scheduled_at}
- 2 On send time: Kafka: announcement.send with resolved recipient list

3 Consumer creates in-app notifications (INSERT notifications per user)

4 Email consumer sends bulk email

5 INSERT audit_trail {ANNOUNCEMENT_SENT, recipient_count, scope}

US-ADM-NOT-004

As a LMS Administrator, I want to configure escalation chains for overdue training — notify managers, then directors, then QA, so that overdue training does not go unaddressed because the system automatically escalates up the management chain

Acceptance Criteria

Escalation chain: employee (Day 1 overdue) → direct manager (Day 3) → dept head (Day 7) → site QA (Day 14)

Configurable thresholds per escalation level

Escalation stops when enrollment is completed or cancelled

Escalation events logged in audit trail

Event Workflow

1 Scheduled job: find all enrollments WHERE due_date < NOW() AND status != COMPLETED

2 For each: calculate days_overdue; select escalation recipients based on threshold

3 Kafka: escalation.notification → notifies appropriate level

4 INSERT escalation_events {enrollment_id, level, sent_to, sent_at}

US-ADM-NOT-005

As a LMS Administrator, I want to view notification delivery logs and track open rates, so that I can verify that reminders were sent and identify delivery failures that need manual follow-up

Acceptance Criteria

Log: notification type, recipient, channel, sent_at, delivered, opened (for email)
Filter by: type, recipient, date range, status (sent/failed/bounced)
Resend failed notifications
Event Workflow
1 SELECT notification_logs with filters
2 Email open tracking via pixel (optional, configurable)
3 Resend: re-publish to Kafka; UPDATE notification_logs SET resent=true
US-ADM-NOT-006
As a LMS Administrator, I want to configure notification preferences per user type — frequency, channel, and opt-out rules, so that employees are not overwhelmed by notifications while still receiving mandatory compliance reminders
Acceptance Criteria
User-level preferences: channel (email/in-app), frequency (immediate/daily digest/weekly)
Admin can set minimum mandatory frequency per notification type (overrides user preference)
Mandatory notification types cannot be opted out (21 CFR 11 compliance events)
Event Workflow
1 User updates preferences → UPDATE user_notification_preferences
2 Notification service checks user preference before sending
3 Mandatory types bypass user preference check

12 — Analytics & Business Intelligence

Covers: training analytics dashboards, learner engagement metrics, content effectiveness, predictive compliance forecasting, and custom report builder.

US-ADM-ANL-001

As a LMS Administrator, I want to view a training analytics dashboard with learner engagement and content effectiveness metrics, so that I can identify which courses have low engagement or high failure rates and prioritise content improvements

Acceptance Criteria

Metrics: avg time-to-completion per course, avg score, completion rate, failure rate, re-attempt rate

Content effectiveness: pass rate trend after content updates

Learner engagement: avg sessions per enrollment, dropout rate (abandoned sessions)

Drill down: per course → per department → per individual learner

Event Workflow

- 1 Aggregation queries over training_sessions, assessment_attempts
- 2 Cached analytics views materialised nightly
- 3 On-demand refresh available for urgent reporting needs

US-ADM-ANL-002

As a LMS Administrator, I want to build custom reports using a drag-and-drop report builder, so that I can create and save ad-hoc reports for specific queries without requiring developer support

Acceptance Criteria

Fields available: all user, enrollment, session, assessment, certificate, audit fields

Operations: filter, group by, aggregate (count, avg, sum, min, max)

Saved reports: name, visibility (private/shared)

Schedule: saved report can be scheduled for periodic generation

Event Workflow

- 1 Admin builds report → stores as report_definition JSONB
- 2 On execute: backend translates definition to SQL query
- 3 Results rendered in table; exportable
- 4 INSERT audit_trail {CUSTOM_REPORT_EXECUTED, report_id}

US-ADM-ANL-003

As a LMS Administrator, I want to view predictive compliance forecasting: which departments are at risk of falling below 80% compliance in the next 30 days, so that I can take preventive action before a compliance dip occurs rather than reacting after the fact

Acceptance Criteria

Model: based on current due dates, in-progress rates, historical completion pace

Output: department × predicted compliance % in 30/60/90 days

Risk flags: departments predicted to drop below threshold

Suggested actions: batch reminder, deadline extension, resource allocation

Event Workflow

- 1 Nightly job: for each department, model completion probability per enrollment
- 2 Aggregate to department-level predicted compliance
- 3 Store in compliance_forecasts {department_id, horizon_days, predicted_pct, calculated_at}

US-ADM-ANL-004

As a LMS Administrator, I want to view a training ROI and cost-per-completion analysis, so that senior leadership can see the value of the LMS investment and I can justify budget requests with data

Acceptance Criteria

Metrics: total completions per month, completions per admin FTE, time saved vs manual tracking estimate

Course-level: avg time-to-complete, estimated cost (content_cost / completions)

Trend: completion volume year-over-year

Event Workflow

- 1 Aggregation over training_sessions.time_spent_seconds, enrollment_records counts
- 2 Cost fields entered manually by admin per course

US-ADM-ANL-005

As a LMS Administrator, I want to view a benchmark comparison between sites or departments on key training metrics, so that site directors can see how their site compares to others and identify best practices to share across the organisation

Acceptance Criteria

Benchmarks: compliance %, overdue rate, avg completion time, pass rate

Comparison: site A vs site B vs organisation average

Ranking: league table of departments by compliance % (opt-in for site directors)

Event Workflow

- 1 Aggregation GROUP BY department/site

2 Organisation average calculated as weighted mean

3 Rankings computed server-side and cached

13 — System Configuration & Integrations

Covers: system settings, portal branding, HR system integration, Kafka configuration,

API key management, SCORM API settings, validation documentation, and system health.

Regulatory basis: 21 CFR 11.30 — open systems; GAMP 5 §10 — change management.

US-ADM-SYS-001 13 — System Configuration & Integrations

As a **System Administrator**, I want to configure portal-wide settings: organisation name, logo, timezone, date format, and language defaults, so that the LMS presents a consistent corporate identity and all timestamps are in the correct timezone for our primary regulatory jurisdiction

Acceptance Criteria

- ✓ Settings: org_name, logo_s3_key, primary_timezone, date_format, default_language
- ✓ Date format default: MMM DD, YYYY (e.g. "Mar 21, 2026") — non-negotiable for ALCOA+ clarity
- ✓ Timezone: stored as IANA timezone; all audit timestamps stored in UTC, displayed in local TZ
- ✓ All changes audit-logged

Event Workflow

- 1 Admin updates settings → UPDATE system_config
- 2 INSERT audit_trail (SYSTEM_CONFIG_UPDATED, old_value, new_value)
- 3 Frontend reads config on load; re-render triggered on change

US-ADM-SYS-002 13 — System Configuration & Integrations

As a **System Administrator**, I want to configure and manage the HR system integration (SAP / Workday / Oracle HCM), so that employee master data is always synchronised so the LMS reflects current headcount, departments, and job titles without manual data entry

Acceptance Criteria

- ✓ Integration: REST API or SFTP file-based
- ✓ Sync schedule: configurable (daily full sync, hourly delta sync)
- ✓ Field mapping: HR fields → LMS fields (configurable per HR system)
- ✓ Conflict resolution: HR system is authoritative for personal data; LMS authoritative for training data
- ✓ Sync logs: full history of every sync run with row-level changes

Event Workflow

- 1 Admin configures HR connection → UPDATE hr_integration_config {endpoint, auth, field_mapping JSONB}
- 2 Scheduled job: pulls HR delta → processes each change (NEW/MODIFIED/TERMINATED employee)
- 3 INSERT hr_sync_logs {sync_at, records_processed, changes_applied JSONB}
- 4 INSERT audit_trail {HR_SYNC_COMPLETED, records_count}

US-ADM-SYS-003 13 — System Configuration & Integrations

As a **System Administrator**, I want to manage API keys for external integrations (ERP, BI tools, badge systems), so that external systems can securely consume LMS data via a governed API without sharing admin credentials

Acceptance Criteria

- ✓ API keys: name, created_by, scopes (read:enrollments, read:certificates, etc.), expiry
- ✓ Keys are hashed on storage (shown once on creation)
- ✓ Rate limiting configurable per key
- ✓ Revoke key immediately on security concern

Event Workflow

- 1 Admin creates key → generates cryptographically random 32-byte key

2 INSERT api_keys (name_hash, key_hash, scopes, expiry)

3 Key shown once to admin (plaintext)

4 Revoke: UPDATE api_keys SET revoked=true, revoked_at

5 INSERT audit_trail (API_KEY_CREATED / REVOKED)

US-ADM-SYS-004 13 — System Configuration & Integrations

As a **System Administrator**, I want to **configure the Kafka event bus topics and consumer groups for reliability**, so that all asynchronous events (notifications, audit writes, HR sync) are reliably delivered with configurable retry and dead-letter handling

Acceptance Criteria

✓ Topics: configurable retention, partitions, replication factor

✓ Dead-letter queue: failed events after N retries → dead_letter topic

✓ Admin UI: view DLQ events, retry or discard

✓ Kafka health: consumer lag monitoring dashboard

Event Workflow

1 Admin views Kafka health in System → Message Bus

2 DLQ review: SELECT dead_letter_events with filters

3 Retry: re-publish event to original topic

4 INSERT audit_trail (DLO_EVENT_RETRIED / DISCARDED, event_id)

US-ADM-SYS-005 13 — System Configuration & Integrations

As a **System Administrator**, I want to view system health dashboard: database connections, S3 storage, Kafka lag, and API response times, so that I can proactively detect and resolve infrastructure issues before they impact learners or compliance deadlines

Acceptance Criteria

- ✓ Metrics: DB connection pool usage, query latency p95, S3 storage used, Kafka consumer lag per topic, API error rate
- ✓ Alerts: configurable thresholds with in-app and email alerts
- ✓ Historical: 30-day trend for all metrics

Event Workflow

- 1 Metrics collected by background health check service every 60 seconds
- 2 Stored in system_metrics time-series table
- 3 Alerts: INSERT system_alerts if threshold exceeded; Kafka: system.alert

US-ADM-SYS-006 13 — System Configuration & Integrations

As a **System Administrator**, I want to manage the system validation documentation required for GAMP 5 and 21 CFR Part 11 computerised system validation, so that the LMS is accompanied by all required validation documents (URS, FS, DQ, IQ, OQ, PQ) that regulators expect for a computerised system used in a GMP environment

Acceptance Criteria

- ✓ Documents managed: URS, Functional Spec, Design Qualification, IQ, OQ, PQ, Validation Summary
- ✓ Each document: version, status, approved_by, approval_date, linked_system_version
- ✓ Validation documents stored in read-only vault (immutable once approved)
- ✓ Re-validation trigger: system flags validation docs for review when software version changes

Event Workflow

- 1 Admin uploads validation document --> INSERT validation_documents (doc_type, version, s3_key, file_hash)

- 2
- On software update: UPDATE validation_documents SET needs_review=true for affected docs
- 3
- INSERT audit_trail (VALIDATION_DOCUMENT_UPLOADED / APPROVED)

US-ADM-SYS-007

13 — System Configuration & Integrations

As a **System Administrator**, I want to **configure backup, disaster recovery settings, and test restore procedures**, so that in the event of a system failure, we can restore full training records within our RTO/RPO targets without any data loss

Acceptance Criteria

- ✓ Backup: daily full + hourly incremental to S3 cross-region
- ✓ RPO target: ≤ 1 hour; RTO target: ≤ 4 hours
- ✓ Admin can trigger manual backup before major changes
- ✓ Restore test: quarterly scheduled restore drill to staging environment
- ✓ All backup/restore operations logged with file hashes

Event Workflow

- 1
- Backup job: pg_dump → S3 encrypted; UPDATE backup_logs (backup_s3_key, file_hash, size_bytes)
- 2
- Restore test: restore to staging; run integrity checks
- 3
- INSERT audit_trail (BACKUP_COMPLETED / RESTORE_TEST_COMPLETED, result)

14 — Data Governance & Archival

US-ADM-DGV-001 14 — Data Governance & Archival

As a **Data Governance Administrator**, I want to configure and enforce data retention policies per record type, so that

Acceptance Criteria

✓

✓

✓

✓

Event Workflow

1

2

3

4

US-ADM-DGV-002 14 — Data Governance & Archival

As a **Data Governance Administrator**, I want to handle a GDPR right-to-erasure request while preserving all regulatory-required records, so that

Acceptance Criteria

✓

✓

✓

✓

Event Workflow

1

2

3

4

5

US-ADM-DGV-003 14 — Data Governance & Archival

As a **Data Governance Administrator**, I want to export a complete data package for a departing employee (data portability), so that

Acceptance Criteria

✓

✓

✓

✓

Event Workflow

1

2

3

US-ADM-DGV-004

14 — Data Governance & Archival

As a **Data Governance Administrator**, I want to manage data classification labels on all stored records, so that

Acceptance Criteria

✓

✓

✓

Event Workflow

1

2

3

US-ADM-DGV-005

14 — Data Governance & Archival

As a **Data Governance Administrator**, I want to generate a data lineage report showing how training records flow from creation to archival, so that

Acceptance Criteria

✓	
✓	
✓	
Event Workflow	
1	
2	
3	

Appendix A — Complete Admin Audit Event Matrix

Every state-changing admin action emits an immutable audit_trail record. The following matrix lists all 75+ admin-triggered audit events across all 14 modules.

Event Code	Module	Trigger	21 CFR / GMP Ref
USER_CREATED	User Management	New user account created by admin	11.10(g)
BULK_USER_IMPORT	User Management	CSV or HR sync import completed	11.10(g)
USER_ROLE_ASSIGNED	User Management	Role added to user	11.10(g)
USER_DEACTIVATED	User Management	Account soft-deleted	11.10(g)
USER_REACTIVATED	User Management	Account restored	11.10(g)
SSO_CONFIG_UPDATED	User Management	SSO/LDAP settings changed	11.300
SECURITY_POLICY_UPDATED	User Management	Password/MFA policy changed	11.300
ACCESS_REVIEW_COMPLETED	User Management	Quarterly access review closed	11.10(g)
COURSE_CREATED	Course Management	New course record created	Annex 11 §10
SCORM_UPLOADED	Course Management	SCORM package uploaded	Annex 11 §10
COURSE_APPROVED	Course Management	Course approved through workflow	Annex 11 §10
COURSE_RETIRED	Course Management	Course moved to RETIRED status	Annex 11 §10
COURSE_VERSION_SUPERSEDED	Course Management	Old version superseded by new	Annex 11 §10
ENROLLMENT_CREATED	Enrollment	Single enrollment created by admin	11.10(e)
BULK_ENROLLMENT_IMPORTED	Enrollment	CSV bulk enrollment committed	11.10(e)
AUTO_ENROLLMENT_TRIGGERED	Enrollment	Rule-based auto-enrollment fired	11.10(e)
ENROLLMENT_MODIFIED	Enrollment	Due date or course version changed	11.10(e), ALCOA+
ENROLLMENT_CANCELLED	Enrollment	Enrollment cancelled with reason	11.10(e)
RE_ENROLLMENT_GRANTED	Enrollment	Failed learner given new attempt	11.10(e)
TRANSCRIPT_EXPORTED	Enrollment	Employee transcript exported	11.10(e)

Event Code	Module	Trigger	21 CFR / GMP Ref
BATCH_CREATED	Batch Management	New batch record saved as DRAFT	Annex 11 §4
BATCH_ACTIVATED	Batch Management	Batch activated; enrollments created	Annex 11 §4
BATCH_MEMBER_ADDED	Batch Management	Employee added to active batch	Annex 11 §4
BATCH_MEMBER_REMOVED	Batch Management	Employee removed from batch	Annex 11 §4
BATCH_REPORT_GENERATED	Batch Management	Compliance report generated for batch	Annex 11 §4
JOB_SPEC_PUBLISHED	Job Specifications	Job spec approved and published	ICH Q10
JOB_SPEC_ASSIGNED_TO_USER	Job Specifications	Spec assigned to employee; enrollments created	11.10(e)
TRAINING_MATRIX_UPDATED	Job Specifications	Matrix cell value changed	ICH Q10
GAP_ANALYSIS_RUN	Job Specifications	Compliance gap analysis executed	Annex 11 §9
JOB_SPEC_NEW_VERSION	Job Specifications	New version of spec published	ICH Q10
ASSESSMENT_CREATED	Assessment Management	New assessment configured	11.10(e)
ASSESSMENT_OVERRIDE_APPLIED	Assessment Management	Result manually overridden (two-person)	11.10(e), ALCOA+
ASSESSMENT_MIN_TIME_VIOLATION	Assessment Management	Submission rejected: min time not met	11.10(e)
CERTIFICATE_ISSUED	Certificate Management	Certificate auto-issued on completion	11.50
CERTIFICATE_MANUALLY_ISSUED	Certificate Management	Certificate manually issued by admin	11.50
CERTIFICATE_REVOKED	Certificate Management	Certificate revoked (two-person rule)	11.50
CERTIFICATE_REGISTER_EXPORTED	Certificate Management	Bulk certificate register exported	11.10(e)
DOCUMENT_UPLOADED	Document Management	SOP/policy PDF uploaded	Annex 11 §14
DOCUMENT_APPROVED	Document Management	Document approved through workflow	Annex 11 §14
DOCUMENT_ASSIGNED	Document Management	Document assigned to department	Annex 11 §14
DOCUMENT_SUPERSEDED	Document Management	Old version superseded	Annex 11 §14

Event Code	Module	Trigger	21 CFR / GMP Ref
DOCUMENT_ARCHIVED	Document Management	Document archived per retention policy	11.10(c)
GAP_REPORT_GENERATED	Compliance Reporting	Gap report executed	Annex 11 §9
REGULATORY_REPORT_GENERATED	Compliance Reporting	Inspection-ready report generated with e-sig	11.50
TRACEABILITY_MATRIX_EXPORTED	Compliance Reporting	SOP-training traceability exported	Annex 11 §9
COMPLIANCE_REPORT_GENERATED	Compliance Reporting	Per-role compliance report generated	Annex 11 §9
AUDIT_TRAIL_SEARCHED	Audit Trail	Admin searched audit trail	11.10(e)
INTEGRITY_CHECK_RUN	Audit Trail	HMAC chain integrity verified	11.10(e)
CAPA_RAISED	CAPA Management	CAPA record created	ICH Q10
CAPA_CLOSED	CAPA Management	CAPA closed with evidence	ICH Q10
ESIGNATURE_INTEGRITY_CHECKED	Audit Trail	E-signature hash verified	11.50
INSPECTION_PACKAGE_GENERATED	Audit Trail	Regulatory dossier assembled with e-sig	11.50
NOTIFICATION_TEMPLATE_UPDATED	Notifications	Email/in-app template edited	GxP
ANNOUNCEMENT_SENT	Notifications	Broadcast sent to user group	GxP
SYSTEM_CONFIG_UPDATED	System Config	System-wide settings changed	11.30
HR_SYNC_COMPLETED	System Config	HR integration sync run	11.10(g)
API_KEY_CREATED	System Config	API key provisioned for integration	11.30
API_KEY_REVOKED	System Config	API key revoked	11.30
BACKUP_COMPLETED	System Config	Database backup completed	11.10(c)
VALIDATION_DOCUMENT_UPLOADED	System Config	GAMP 5 validation document stored	GAMP 5 §10
RECORDS_ARCHIVED	Data Governance	Records moved to archive tier	11.10(c), EU Annex 11 §17
GDPR_ERASURE_EXECUTED	Data Governance	Personal data pseudonymised (DSAR)	GDPR Art. 17
DATA_EXPORT_GENERATED	Data Governance	Employee data package exported	GDPR Art. 20

Appendix B — Admin Portal DB Write Surface

Every table that receives INSERT or UPDATE operations triggered by admin actions.

Table	Admin Operations	Module(s)	Regulatory Importance
users	INSERT, UPDATE (active, roles, profile), soft-delete	User Mgmt	21 CFR 11.10(g)
user_roles	INSERT, UPDATE, DELETE (cascade)	User Mgmt	21 CFR 11.10(g)
role_history	INSERT (append-only)	User Mgmt	Audit trail for role changes
user_sessions	DELETE (on deactivation/revocation)	User Mgmt	21 CFR 11.300
access_reviews	INSERT	User Mgmt	Privileged access evidence
departments	INSERT, UPDATE	User Mgmt	Org structure
courses	INSERT, UPDATE (status, metadata)	Course Mgmt	GMP Annex 11 §10
scorm_packages	INSERT	Course Mgmt	Content integrity
course_approval_workflow	INSERT, UPDATE	Course Mgmt	GAMP 5
course_sop_links	INSERT, DELETE	Course Mgmt	Traceability
enrollment_records	INSERT, UPDATE (status, due_date, cancel)	Enrollment	21 CFR 11.10(e)
import_jobs	INSERT	Enrollment	Bulk import record
enrollment_rules	INSERT, UPDATE	Enrollment	Auto-enrollment config
training_batches	INSERT, UPDATE (status, activate)	Batch Mgmt	GMP Annex 11 §4
batch_courses	INSERT, DELETE	Batch Mgmt	Batch configuration
batch_members	INSERT	Batch Mgmt	Roster record
job_specifications	INSERT, UPDATE (version, status)	Job Spec	ICH Q10
job_spec_courses	INSERT, UPDATE, DELETE	Job Spec	Training matrix
job_spec_versions	INSERT	Job Spec	Change control
assessment_questions	INSERT, UPDATE (status)	Assessment Mgmt	21 CFR 11.10(e)
assessments	INSERT, UPDATE	Assessment Mgmt	21 CFR 11.10(e)

Table	Admin Operations	Module(s)	Regulatory Importance
assessment_overrides	INSERT	Assessment Mgmt	Exception management
certificates	INSERT (manual), UPDATE (revoke)	Cert Mgmt	21 CFR 11.50
certificate_templates	INSERT, UPDATE	Cert Mgmt	Template management
sop_versions	INSERT, UPDATE (status, supersede, archive)	Doc Mgmt	GMP Annex 11 §14
department_sop_assignments	INSERT, UPDATE	Doc Mgmt	Distribution control
document_workflow_instances	INSERT, UPDATE	Doc Mgmt	Approval evidence
capa_records	INSERT, UPDATE (status, close)	CAPA Mgmt	ICH Q10
notification_templates	UPDATE	Notifications	Communication config
reminder_rules	INSERT, UPDATE	Notifications	Escalation config
announcements	INSERT	Notifications	Broadcast management
system_config	UPDATE	System Config	21 CFR 11.30
hr_integration_config	UPDATE	System Config	HR sync config
api_keys	INSERT, UPDATE (revoke)	System Config	Access control
scheduled_reports	INSERT, UPDATE	Analytics	Reporting automation
audit_trail	INSERT (append-only — EVERY operation)	ALL	21 CFR 11.10(e) — immutable